Data Systems Lab 01

Classification: Engineering / Software Lab Report **Status:** Completed Documentation

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1. Objective & Context Summary

This engineering module demonstrates building a highly performant architecture leveraging decoupled modular components.

2. Implementation Source Code Artifacts

server.ts (ts)

```ts
import app from './app.js';

const PORT = process.env.PORT || 3000;

const server = app.listen(PORT, () => {
  console.log(`=======================================`);
  console.log(`🚀 Lab Report Engine Online`);
  console.log(`📡 Listening on: http://localhost:${PORT}`);
  console.log(`=======================================`);
});

// Implement clean process teardowns to prevent zombie headless browser processes
process.on('SIGTERM', () => {
  console.log('SIGTERM signal received. Shutting down gracefully...');
  server.close(() => {
    console.log('HTTP Server closed.');
    process.exit(0);
  });
});
```

3. Execution Outputs & Operational Logs

```
[SUCCESS] App compiled in 42ms
[INFO] Execution thread finished with exit code 0
```